Homework #3 - Spaceflight Mechanics

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1 Part 1

1.1 Problem

A spacecraft starts with horizontal speed from a direct terrestrial parking orbit (LEO) with a radius of 6700 km and enters a trajectory that takes it towards the Moon. Consider the following simplification hypotheses:

- circular lunar orbit with radius $R_M = D_{em} = 384400\ km$
- lunar orbit, LEO, and Earth-Moon flight all coplanar

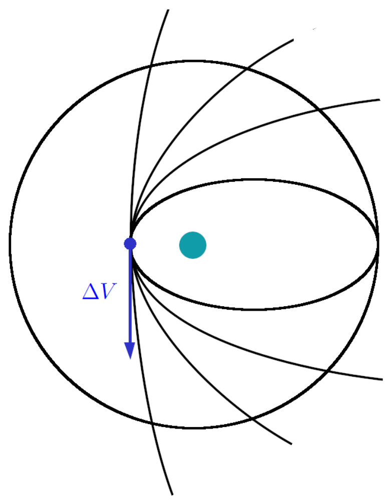
Neglecting the gravitational attraction of the Moon, evaluate and plot the following quantities as the starting speed V_0 varies: semi-axis and eccentricity of the transfer trajectory; the angle swept by the spacecraft and the time needed to intercept the Moon after leaving the LEO; the ratio of the variations of these last two quantities for small V_0 increments, and the spacecraft/Moon angle at the start.

1.2 The Hohmann transfer

The Hohmann manuever can be seen as a limit case for the earth-moon transfer. It is the most economic transfer in terms of ΔV, but it's also the longest route to the Moon. There's some data to begin with.

$$D_{em} = 384400 km$$
$$R_{soi,moon} = 66300\ km$$
$$r_0 = 6700\ km$$

$$a_H = \frac{r_0 + D_{em}}{2} = 195550\ km$$
$$e_H = \frac{D_{em} - r_0}{D_{em} + r_0} = 0.9657$$
$$v_H = \sqrt{\frac{\mu}{a_H}} \cdot \sqrt{\frac{1 + e_H}{1 - e_H}} = 10.8142\ km/s$$



1.3 The other transfers

Let's look now at the spectrum of possibilities to get to the Moon. We can either use ellipses or hyperbolas, but the former are best suited as they're close trajectories and thus have less risk for manned spacecraft. Here are the initial conditions of the spacecraft.

$$r_0 = 6700\ km$$
$$v_0 = [v_H, 12\ km/s]$$
$$\phi_0 = 0\ deg$$

We first need to estimate the semi-major axis and the eccentricity of the elliptical transfers respectively from the vis-viva equation and the angular momentum definition. Then we'll be able to calculate both the angle swept by the spacecraft and the time required to do so using either *Kepler's equation* for ellipses or hyperbola. Lastly, the initial position of the Moon can be deduced from the schematics below.

$$\epsilon = \frac{v_0^2}{2} - \frac{\mu_E}{r_0} = -\frac{\mu_E}{2a_T} \qquad \Rightarrow \qquad a_T = \frac{\mu_E}{\left(\frac{2\mu_E}{r_0} - v_0^2\right)}$$

$$h = p\mu_E = a\mu_E(1-e_T^2) \qquad \Rightarrow \qquad e_T = \sqrt{1 - \frac{(r_0 v_0)^2}{\mu_E a_T}}$$

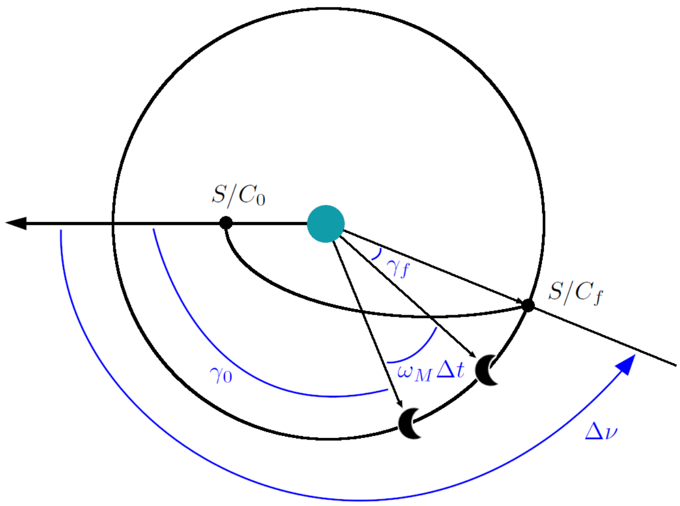
$$p_T = a_T(1-e_T^2)$$

$$\nu_{fin} = \arccos\left(\frac{p_T - D_{em}}{D_{em} e_T}\right)$$
$$\nu_{ini} = 0$$
$$\Delta\nu = \nu_{fin} - \nu_{ini}$$

$$\Delta t = \begin{cases} \sqrt{\frac{a^3}{\mu}}(E_{fin} - e_T \sin E_{fin} - E_{ini} + e_T \sin E_{ini}) & \text{for an ellipse} \\ \sqrt{-\frac{a^3}{\mu}}(e_T \sinh F_{fin} - F_{fin} - e_T \sinh F_{ini} + F_{ini}) & \text{for an hyperbola} \end{cases} \quad \text{where} \quad \begin{cases} \cos E = \frac{e + \cos\nu}{1 + e\cos\nu} \\ \cosh F = \frac{e + \cos\nu}{1 + e\cos\nu} \end{cases}$$

$$\frac{\Delta\nu}{\Delta t} = \frac{\Delta\nu(i+1) - \Delta\nu(i)}{\Delta t(i+1) - \Delta t(i)}$$

$$\gamma_0 = \Delta\nu - \omega_M \Delta t$$



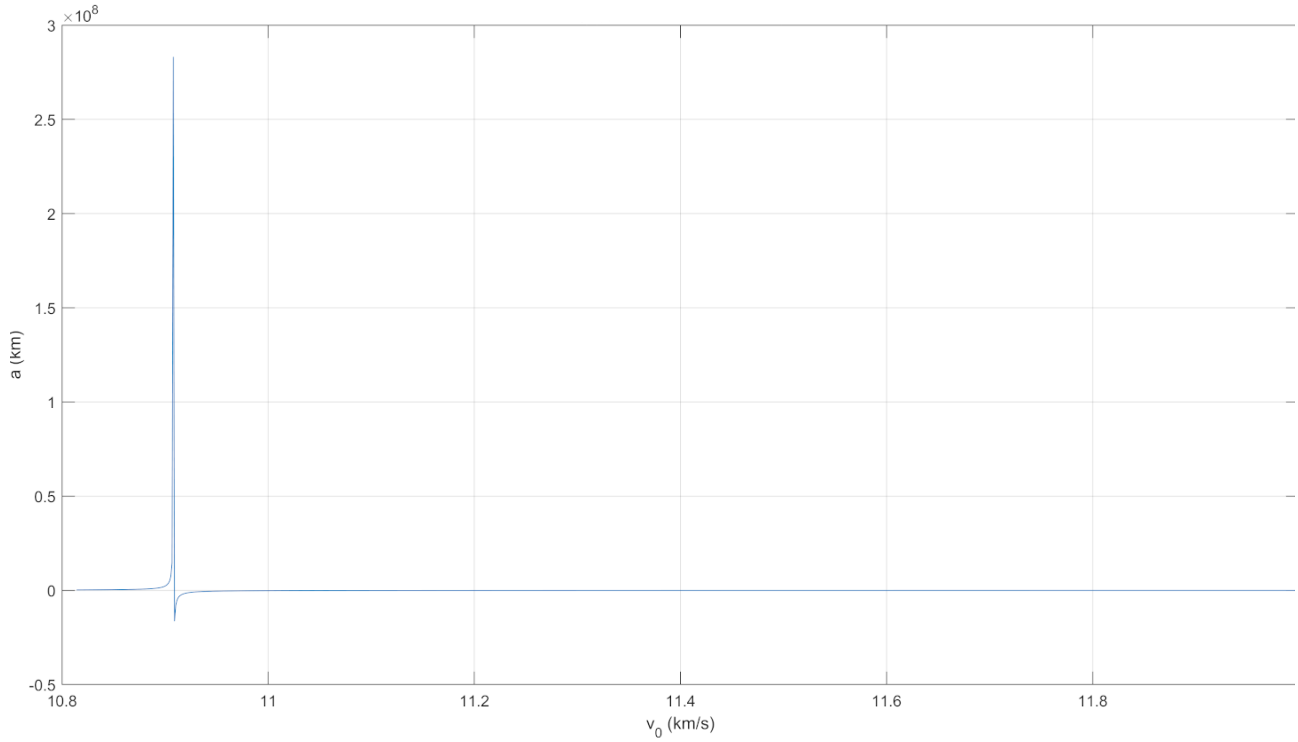


FIGURE 1. Semi-major axis a as v_0 varies

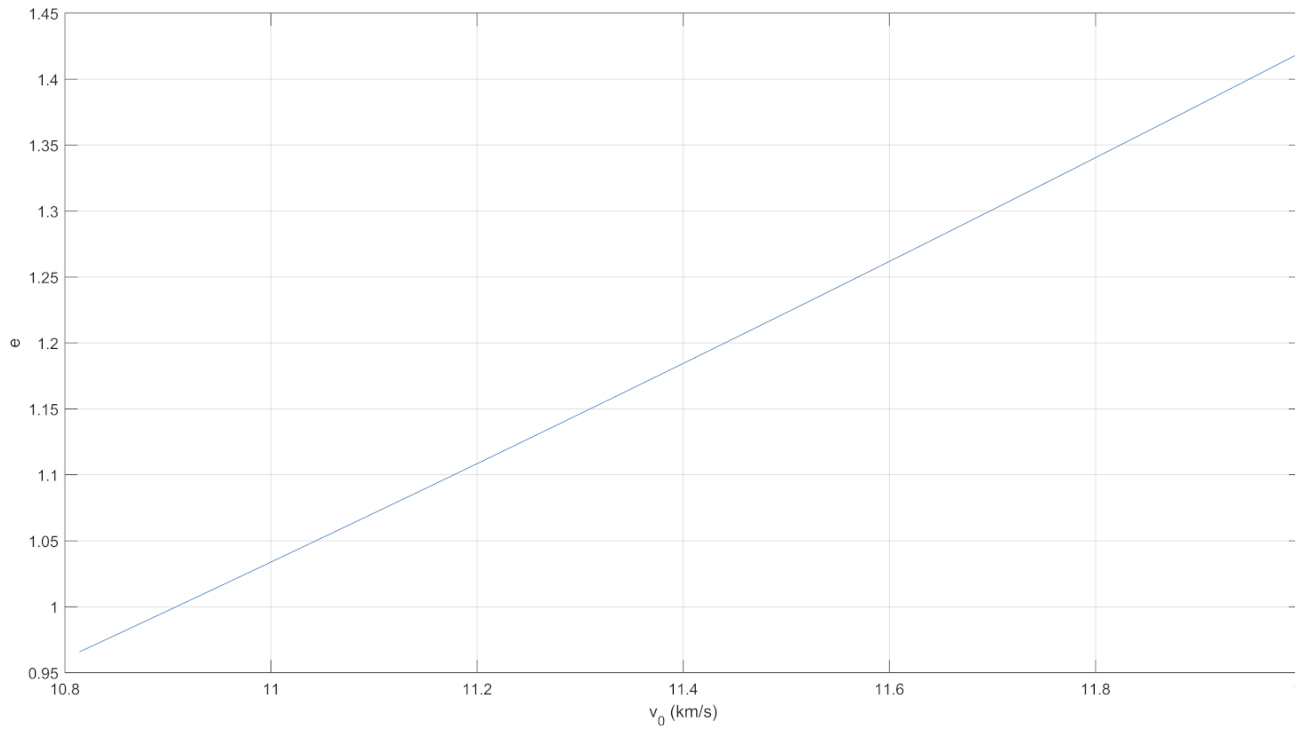


FIGURE 2. Eccentricity e as v_0 varies

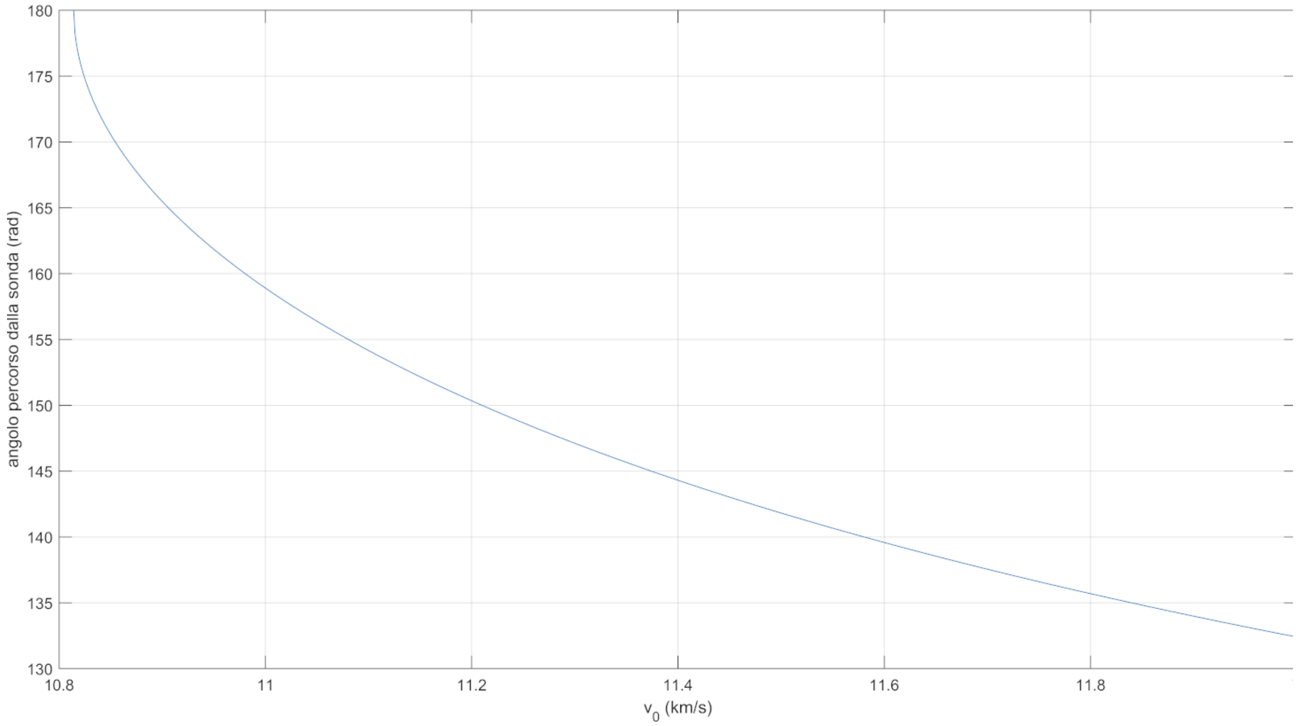


FIGURE 3. Angle $\Delta\nu$ swept by the spacecraft as v_0 varies

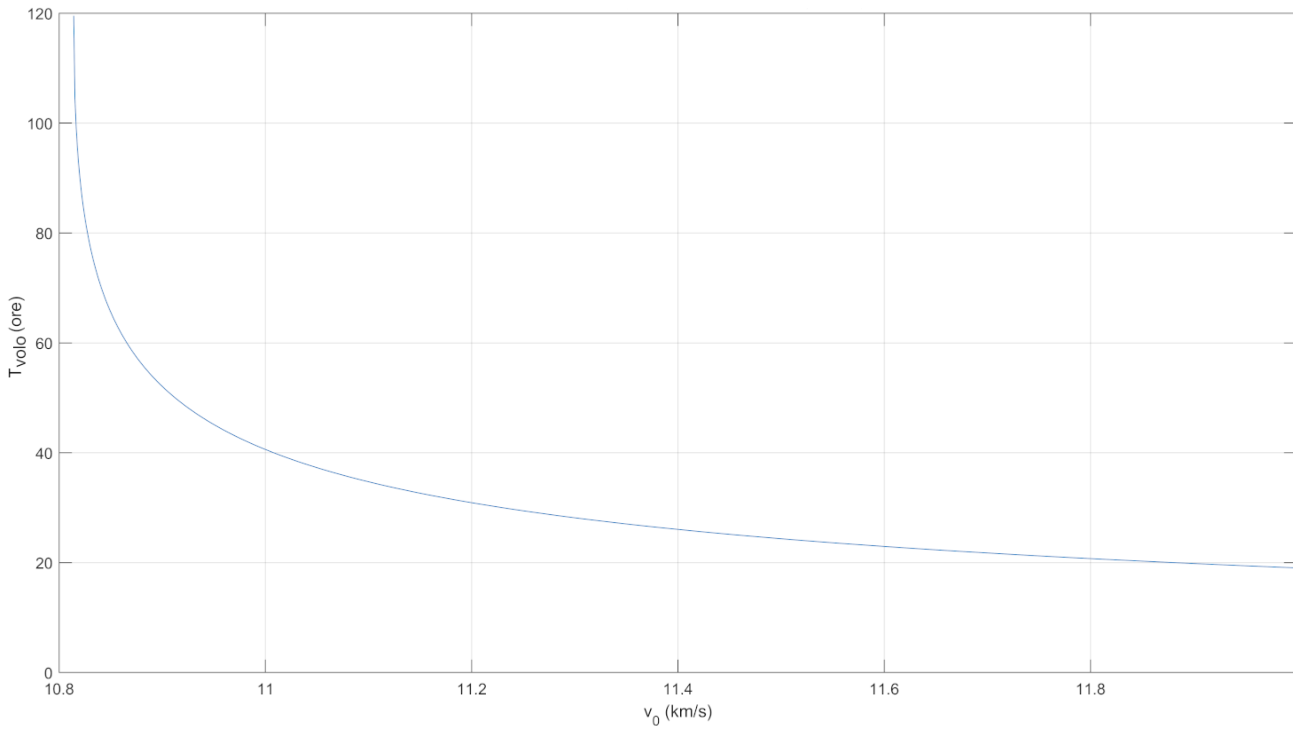


FIGURE 4. Flight time Δt as v_0 varies

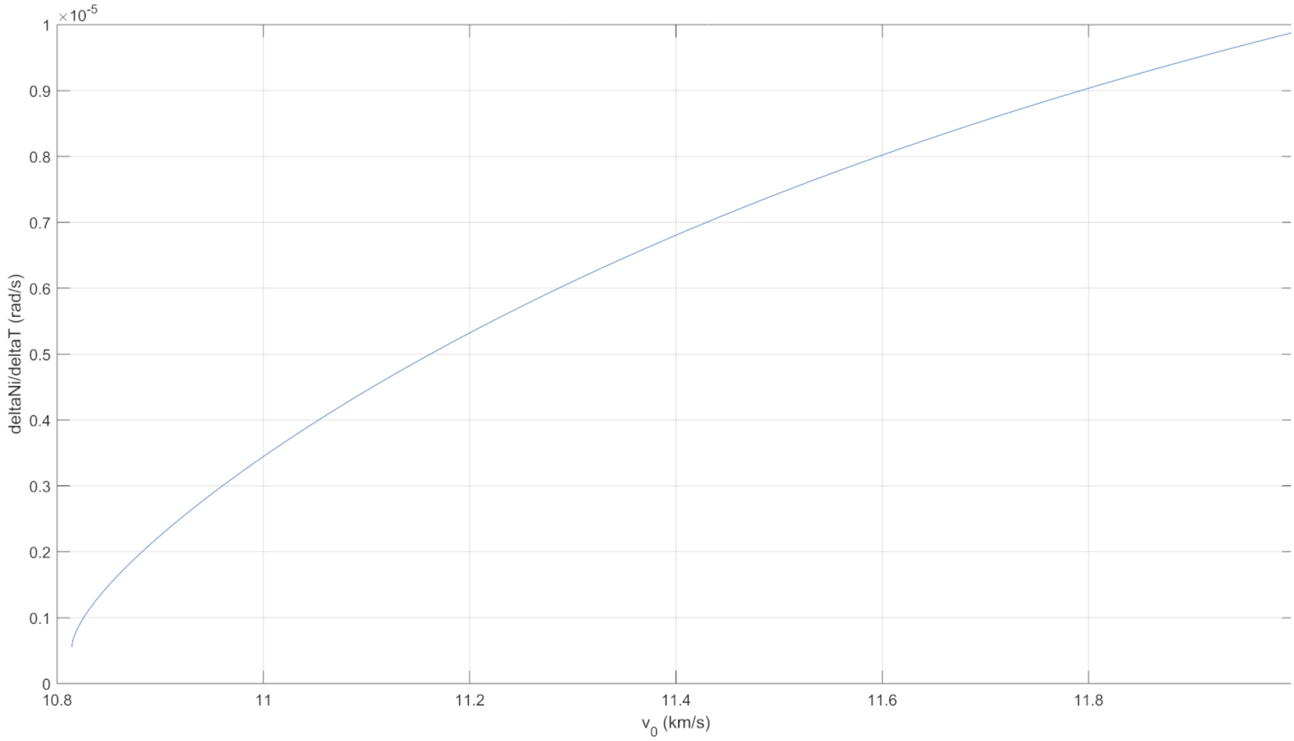


FIGURE 5. Ratio $\frac{\Delta\nu}{\Delta T}$ as v_0 varies

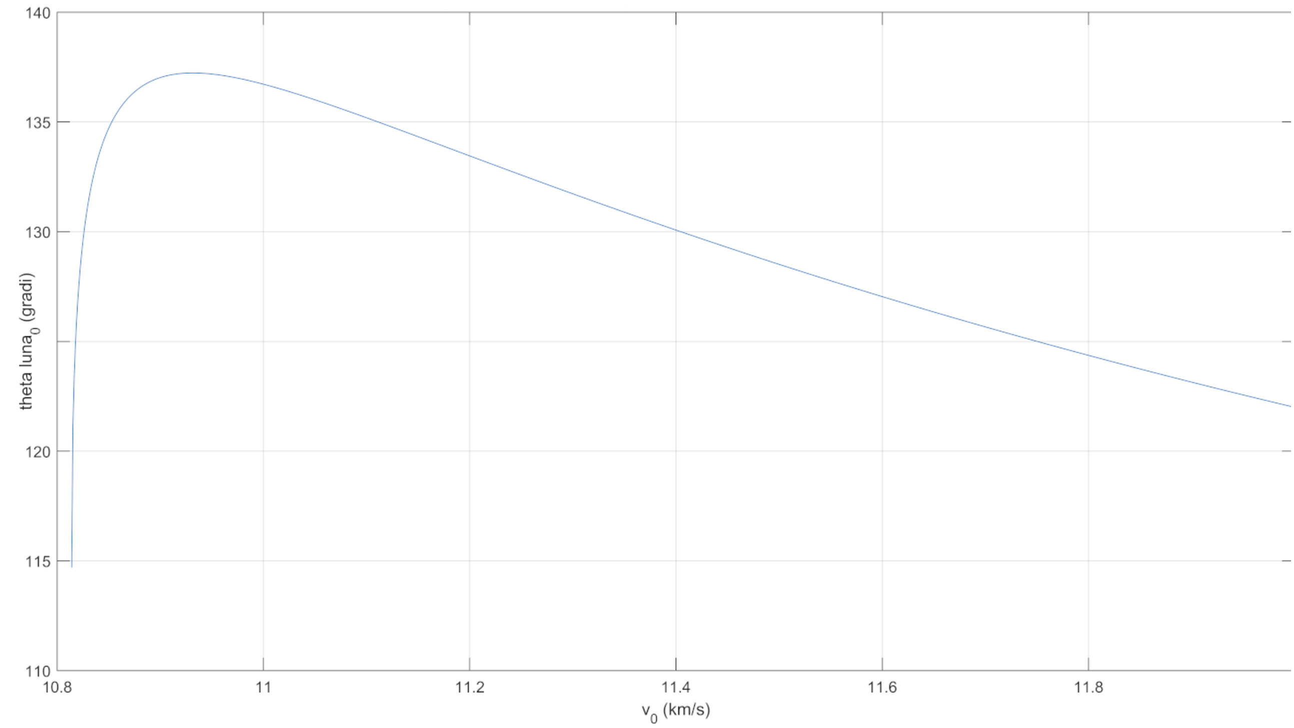


FIGURE 6. Angle spacecraft/Moon γ_0 as v_0 varies

2 Part 2

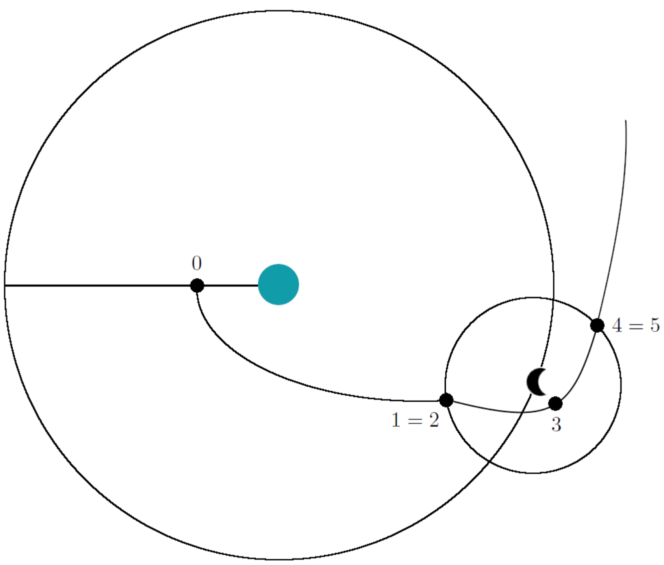
2.1 Problem

Once an appropriate value of V_0 has been chosen, analyze the trajectory of the spacecraft with the patched-conic model, assuming for the Moon a radius of 1738 km and a gravitational parameter μ_M of 4890 km^3/s^2. Also, calculate the ΔV required for insertion into direct and retrograde low lunar orbit (radius of the periselenium between 1850 and 2300 km). In both cases, estimate the conditions of the probe (position and speed and specific energy in the geocentric reference frame) at the exit of the lunar sphere of influence if no engine ignition occurs at the periselenium.

2.2 Patched conic model

The patched conic model will be used from here onwards as it is more accurate than simply neglecting the lunar attraction, while retaining some simplifications from the three body problem. The Moon has now an area, called the *sphere of influence* where its gravitational pull is stronger than Earth's. In the first part of the trajectory we'll neglect the lunar attraction, while in the second part we'll forget about the Earth.
For the periselenium height we choose $r_{per} = 2000$ km. The following image depicts the trajectory of the spacecraft as well as represents the prominent points along its route.

- P0 : starting point, geocentric frame
- P1 : entering soi of the Moon, geocentric frame
- P2 : entering soi of the Moon, selenocentric frame
- P3 : periselenium, selenocentric frame
- P4 : exiting soi of the Moon, selenocentric frame
- P5 : exiting soi of the Moon, geocentric frame



2.3 Geocentric leg

That's the trajectory from P0 to P1 in the geocentric frame. We already have the conditions about r_0 and ϕ_0, while we had to select the speed v_0 as to guarantee an elliptical trajectory for safety concerns. From there, we can calculate the specific mechanical energy ϵ, the specific angular momentum h and the geometric characteristics of the geocentric leg. Creating a vector for λ_1 provides us a series of results for each parameter: we'll select these later to ensure capture in lunar orbit in the two cases.

- **P0**

$$r_0 = 6700\ km$$
$$v_0 = 10.87\ km/s$$
$$\phi_0 = 0\ deg$$

$$\epsilon_{01} = \frac{v_0^2}{2} - \frac{\mu_E}{r_0}$$
$$h_{01} = v_0 r_0 \cos\phi_0$$
$$a_{01} = -\frac{\mu_E}{2\epsilon_{01}}$$
$$p_{01} = \frac{h_{01}^2}{\mu_E}$$
$$e_{01} = \sqrt{1 - \frac{p_{01}}{a_{01}}}$$
$$\nu_0 = 0$$
$$E_0 = 0$$
$$M_0 = 0$$

- **P1**

$$\lambda_1 = [0, \ldots, \pi]$$

$$r_1 = \sqrt{D_{em}^2 + R_{soi,m}^2 + 2D_{em}R_{soi,m}\cos\lambda_1}$$
$$v_1 = \sqrt{2\epsilon_{01} + \frac{2\mu_E}{r_1}}$$
$$\phi_1 = \arccos\left(\frac{h_{01}}{r_1 v_1}\right)$$

$$\nu_1 = \arccos\left(\frac{p_{01} - r_1}{r_1 e_{01}}\right)$$
$$E_1 = 2\arctan\left(\sqrt{\frac{1-e}{1+e}}\tan\frac{\nu_1}{2}\right)$$
$$M_1 = E_1 - e_{01}\sin E_1$$
$$\gamma_1 = -\arcsin\left(\frac{R_{soi,m}}{r_1}\sin\lambda_1\right)$$
$$\Delta t_{01} = (M_1 - M_0)\sqrt{\frac{a_{01}^3}{\mu_E}}$$

2.4 Selenocentric leg

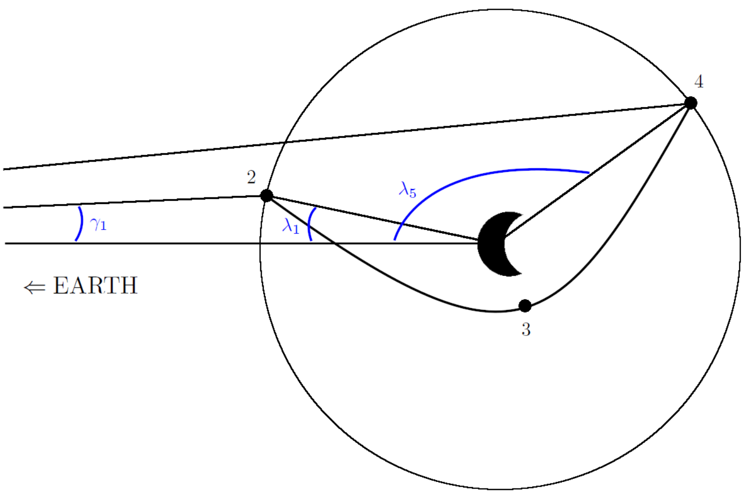
That's the trajectory going from P2 to P4, passing through P3 in the selenocentric frame.

- **P2**

$$
\begin{aligned}
&r_2 = R_{soi,m}\\
&v_2 = \sqrt{v_1^2 + v_M^2 - 2v_1 v_M \cos(\phi_1 + \gamma_1)}\\
&d\epsilon_2 = \arcsin\left(\frac{v_1}{v_2}\cos(\lambda_1 + \gamma_1 + \phi_1) - \frac{v_M}{v_2}\cos\lambda_1\right)\\
&\epsilon_{234} = \frac{v_2^2}{2} - \frac{\mu_M}{R_{soi,m}}\\
&h_{234} = R_{soi,m} v_2 \sin(d\epsilon_2)\\
&p_{234} = \frac{h_{234}^2}{\mu_M}\\
&e_{234} = \sqrt{1 + 2\epsilon_{234}\frac{h_{234}^2}{\mu_M^2}}\\
&a_{234} = \frac{p_{234}}{(1 - e_{234}^2)}\\
&\nu_2 = -\arccos\left(\frac{p_{234} - R_{soi,m}}{R_{soi,m} e_{234}}\right)\\
&F_2 = 2\,\text{atanh}\left(\sqrt{\frac{e_{234} - 1}{e_{234} + 1}}\tan\frac{\nu_2}{2}\right)\\
&M_2 = e_{234}\sinh F_2 - F_2
\end{aligned}
$$

- **P3**

$$
\begin{aligned}
&r_3 = \frac{p_{234}}{1 + e_{234}}\\
&v_3 = \sqrt{2\left(\epsilon_{234} + \frac{\mu_M}{r_3}\right)}\\
&\phi_3 = 0\\
&\nu_3 = 0
\end{aligned}
$$



Since we used a vector for r_3, we need to make sure to take the values close to $r_{periselenium}$ and use those to estimate the characteristic values.

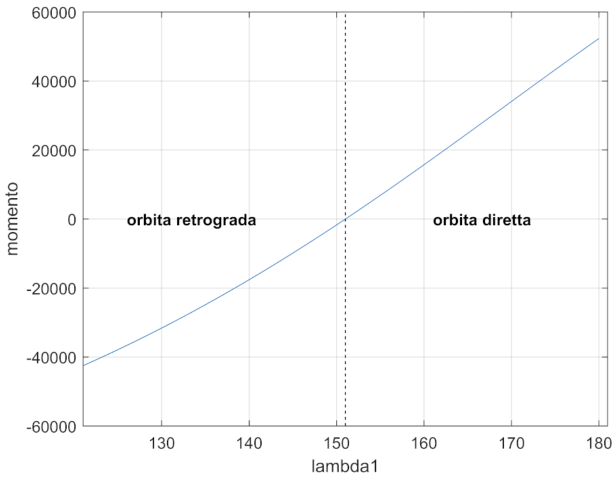


FIGURE 7. λ_1 vs h_{234}

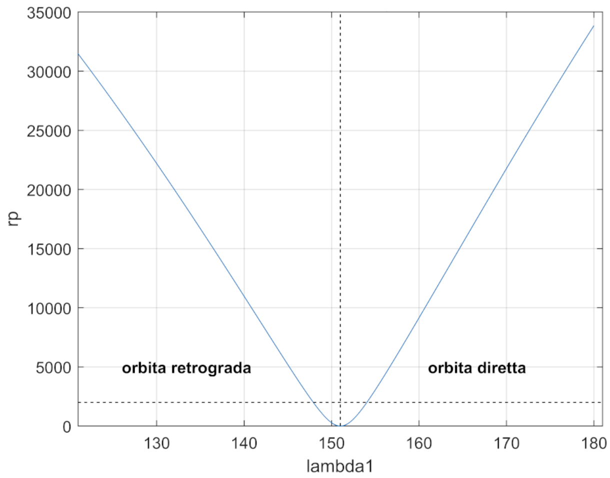


FIGURE 8. λ_1 vs r_{per}

From Fig. 8 we can identify two values of λ_1 that guarantee a periselenium equal to 2000 km. Let's study the case of no ignition at the pericenter, and therefore the departure of the spacecraft from the lunar soi.

$$\begin{cases} \lambda_{1,direct} = 154.04° \\ \lambda_{1,retrograde} = 147.90° \end{cases}$$

- **P4**

$$r_4 = R_{soi,m}$$
$$v_4 = v_2$$
$$d\epsilon_4 = -d\epsilon_2 + pi$$
$$\nu_4 = \nu_2$$
$$F_4 = 2\,\text{atanh}\left(\sqrt{\frac{e_{234}-1}{e_{234}+1}}\tan\frac{\nu_4}{2}\right)$$
$$M_4 = e_{234}\sinh F_4 - F_4$$
$$\Delta t_{24} = (M_4 - M_2)\sqrt{-\frac{a_{234}^3}{\mu_M}}$$

- **P5**

$$\begin{cases} \lambda_{5,direct} = \lambda_{1,direct} - 2\nu_2 = 270.86° \\ \lambda_{5,retrograde} = \lambda_{1,retrograde} + 2\nu_2 = 32.26° \end{cases}$$
$$r_5 = \sqrt{R_{soi,m}^2 + D_{em}^2 + 2R_{soi,m}D_{em}\cos\lambda_5} \Rightarrow \begin{cases} r_{5,direct} = 391050\ km \\ r_{5,retrograde} = 441880\ km \end{cases}$$
$$v_5 = \sqrt{[v_4\cos(d\epsilon_4+\lambda_5)]^2 + [v_4\sin(d\epsilon_4+\lambda_5)+v_M]^2} \Rightarrow \begin{cases} v_{5,direct} = 2.4227\ km/s \\ v_{5,retrograde} = 1.1450\ km/s \end{cases}$$
$$\epsilon_5 = \frac{v_5^2}{2} - \frac{\mu_E}{r_5} \Rightarrow \begin{cases} \epsilon_{5,direct} = 1.9153 \\ \epsilon_{5,retrograde} = -0.2466 \end{cases}$$

2.5 ΔV_{tot} estimation for lunar capture

The total cost of the mission consists of two impulses: the first thrust from the parking orbit to begin the Earth-Moon transfer and the second at the periselenium to slow down. We imagine to capture the spacecraft in an elliptical orbit, with an apogee of 40000 km (still inside the lunar soi).

$$\Delta V_1 = V_0 - \sqrt{\frac{\mu}{r_0}} = 3.1569\ km/s$$

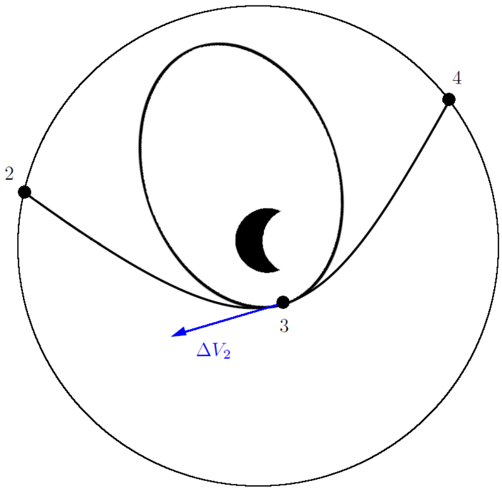
$$a_{m,orbit} = \frac{r_3 + 40000}{2}$$

$$e_{m,orbit} = \frac{40000 - r_3}{40000 + r_3}$$

$$p_{m,orbit} = a_{mo}(1 - e_{mo}^2)$$

$$\Delta V_2 = V_3 - \sqrt{\frac{\mu_M}{a_{mo}}} \cdot \sqrt{\frac{1 + e_{mo}}{1 - e_{mo}}} \Rightarrow \begin{cases} \Delta V_{2,direct} = 0.4343\ km/s \\ \Delta V_{2,retrograde} = 0.4214\ km/s \end{cases}$$

$$\Delta V_{tot} = \Delta V_1 + \Delta V_2 \Rightarrow \begin{cases} \Delta V_{tot,direct} = 3.5911\ km/s \\ \Delta V_{tot,retrograde} = 3.5783\ km/s \end{cases}$$



2.6 Summary

- retrograde trajectory

	0	1	2	3	4	5	
r	6700	330120	66300	2008	66300	441880	$[km]$
v	10.87	1.2596	1.3811	2.5748	1.3811	1.1450	$[km/s]$
ϕ	0	1.3947	—	0	—	1.3006	$[rad]$
ϵ	-0.4141		0.8799			-0.2466	$[km^2/s^2]$
h	72829		-5171			135060	$[km^2/s]$
ν	0	2.9097	-2.1324	0	2.1324	2.7645	$[rad]$
λ	—	2.5814	—	—	—	6.8463	$[rad]$
$d\epsilon$	—	—	-0.0565	—	3.1981	—	$[rad]$
γ	—	-0.1069	—	—	—	—	$[rad]$

TABLE 1. Mission outline data for retrograde trajectory

- direct trajectory

	0	1	2	3	4	5	
r	6700	326080	66300	2008	66300	391050	$[km]$
v	10.87	1.2715	1.4049	2.5877	1.4049	2.4227	$[km/s]$
ϕ	0	1.3942	—	0	—	0.1467	$[rad]$
ϵ	-0.4141		0.9130			1.9153	$[km^2/s^2]$
h	72829		5196			937220	$[km^2/s]$
ν	0	2.9075	-2.1222	0	2.1222	0.1777	$[rad]$
λ	—	2.6885	—	—	—	-1.5558	$[rad]$
$d\epsilon$	—	—	0.0558	—	3.0858	—	$[rad]$
γ	—	-0.0891	—	—	—	—	$[rad]$

TABLE 2. Mission outline data for direct trajectory